

Problem

Use *PSpice* or *MultiSim* to solve the problem in Example 3.4.

Example 3.4:

Find the node voltages in the circuit of Fig. 3.12.

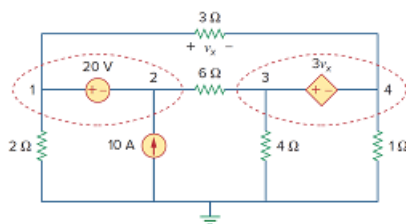


Figure 3.12
For Example 3.4.

Solution:

Nodes 1 and 2 form a supernode; so do nodes 3 and 4. We apply KCL to the two supernodes as in Fig. 3.13(a). At supernode 1-2,

$$i_3 + 10 = i_1 + i_2$$

Expressing this in terms of the node voltages,

$$\frac{v_3 - v_2}{6} + 10 = \frac{v_1 - v_2}{3} + \frac{v_1}{2}$$

or

$$5v_1 + v_2 - v_3 - 2v_4 = 60 \quad (3.4.1)$$

At supernode 3-4,

$$i_1 = i_3 + i_4 + i_5 \Rightarrow \frac{v_1 - v_2}{3} = \frac{v_3 - v_2}{6} + \frac{v_4}{1} + \frac{v_3}{4}$$

or

$$4v_1 + 2v_2 - 5v_3 - 16v_4 = 0 \quad (3.4.2)$$

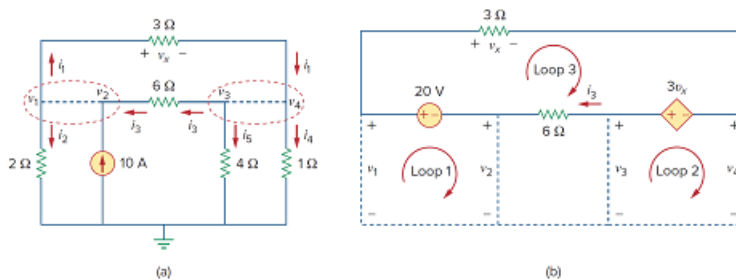


Figure 3.13
Applying: (a) KCL to the two supernodes, (b) KVL to the loops.

We now apply KVL to the branches involving the voltage sources as shown in Fig. 3.13(b). For loop 1,

$$-v_1 + 20 + v_2 = 0 \Rightarrow v_1 - v_2 = 20 \quad (3.4.3)$$

For loop 2,

$$-v_3 + 3v_x + v_4 = 0$$

But $v_x = v_1 - v_4$ so that

$$3v_1 - v_3 - 2v_4 = 0 \quad (3.4.4)$$

For loop 3,

$$v_x - 3v_x + 6i_3 - 20 = 0$$

But $6i_3 = v_3 - v_2$ and $v_x = v_1 - v_4$. Hence,

$$-2v_1 - v_2 + v_3 + 2v_4 = 20 \quad (3.4.5)$$

We need four node voltages, v_1 , v_2 , v_3 , and v_4 , and it requires only four out of the five Eqs. (3.4.1) to (3.4.5) to find them. Although the fifth equation is redundant, it can be used to check results. We can solve Eqs. (3.4.1) to (3.4.4) directly using *MATLAB*. We can eliminate one node voltage so that we solve three simultaneous equations instead of four. From Eq. (3.4.3), $v_2 = v_1 - 20$. Substituting this into Eqs. (3.4.1) and (3.4.2), respectively, gives

$$6v_1 - v_3 - 2v_4 = 80 \quad (3.4.6)$$

and

$$6v_1 - 5v_3 - 16v_4 = 40 \quad (3.4.7)$$

Equations (3.4.4), (3.4.6), and (3.4.7) can be cast in matrix form as

$$\begin{bmatrix} 3 & -1 & -2 \\ 6 & -1 & -2 \\ 6 & -5 & -16 \end{bmatrix} \begin{bmatrix} v_1 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 80 \\ 40 \end{bmatrix}$$

Using Cramer's rule gives

$$\Delta = \begin{vmatrix} 3 & -1 & -2 \\ 6 & -1 & -2 \\ 6 & -5 & -16 \end{vmatrix} = -18, \quad \Delta_1 = \begin{vmatrix} 0 & -1 & -2 \\ 80 & -1 & -2 \\ 40 & -5 & -16 \end{vmatrix} = -480,$$

$$\Delta_3 = \begin{vmatrix} 3 & 0 & -2 \\ 6 & 80 & -2 \\ 6 & 40 & -16 \end{vmatrix} = -3120, \quad \Delta_4 = \begin{vmatrix} 3 & -1 & 0 \\ 6 & -1 & 80 \\ 6 & -5 & 40 \end{vmatrix} = 840$$

Thus, we arrive at the node voltages as

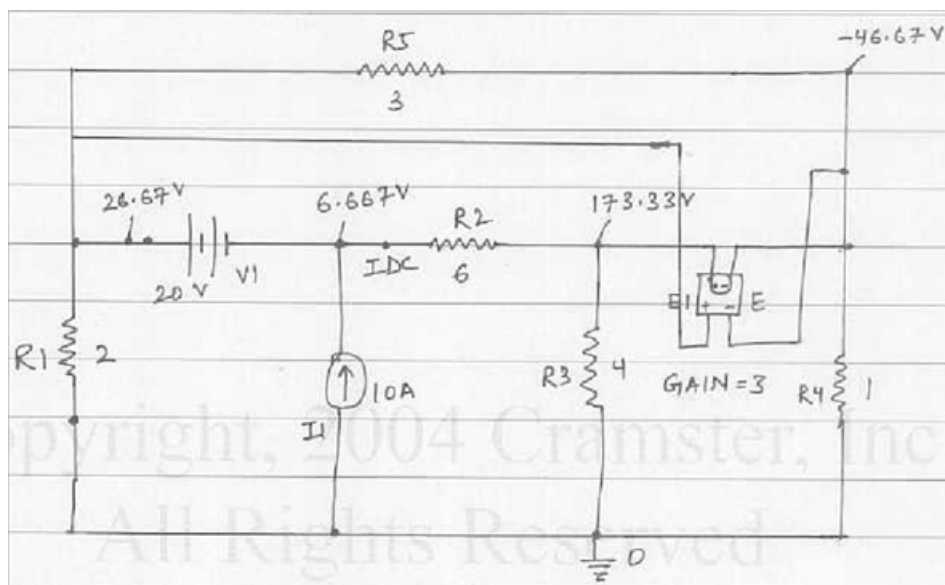
$$v_1 = \frac{\Delta_1}{\Delta} = \frac{-480}{-18} = 26.67 \text{ V}, \quad v_3 = \frac{\Delta_3}{\Delta} = \frac{-3120}{-18} = 173.33 \text{ V},$$

$$v_4 = \frac{\Delta_4}{\Delta} = \frac{840}{-18} = -46.67 \text{ V}$$

and $v_2 = v_1 - 20 = 6.667 \text{ V}$. We have not used Eq. (3.4.5); it can be used to cross-check results.

Step-by-step solution

Step 1 of 1



Clearly,

$$V_1 = 26.67\text{Volts} , \quad V_2 = 6.667\text{Volts} ,$$

$$V_3 = 173.33\text{Volts} \text{ And } V_4 = -46.67\text{Volts} .$$

Which agrees with the results of example 3.4 .

This is the netlist for this circuit.

Schematics Netlist:-

R - R1 0 \$N_0001 2

R - R2 \$ N_0003 \$N_0002 6

R - R3 0 \$N_0002 4

R - R4 0 \$N_0004 1

R - R5 \$ N_0001 \$N_0004 3

I - I1 0 \$N_0003 DC 10 A

V - V1 \$N_0001 \$N_0003 20 V

E - E1 \$ N_0002 \$N_0004 \$ N_0001 \$N_0004 3